HEDGING FOR KURTOSIS

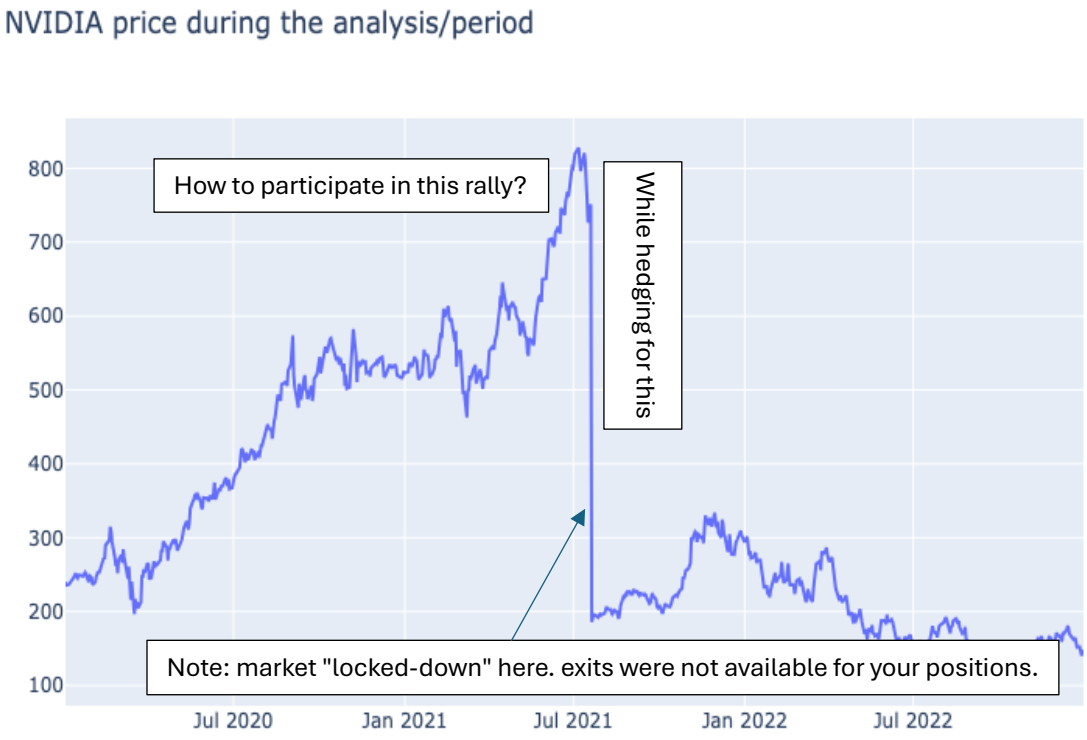
Project Track

Practical Track - Options Trading and Forecasting

Area of Study: Option Strategy for participating in a market rally while hedging for kurtosis.

Problem statement

Sudden market crashes are much more frequent than implied by a log-normal distribution of asset returns. So, when investors participate in the market rally, they need protection against sudden crashes or maybe even a market malfunction. Markets might even "lock-down" temporarily and exiting the position becomes unachievable during such periods. Below image illustrates this.



Main Research Question

This study explores if an options strategy with the following criteria can be created.

- the strategy allows for participation in a market rally.
- the strategy hedges for sharp moves/kurtosis.
- the strategy preserves the value of positions during a temporary market lock-down.
- placed positions cannot be forcibly exited by the market-maker via. a margin call mechanism.
- the strategy can be hard-coded, & thereafter does not need manual monitoring/adjustments until lock-down.

However, note that the strategy does not aim to protect against

- A more permanent lock-down of the market, or
- If the market-maker defaults on the already executed contracts.

Scope

Market freeze is a rare event. So, data related to bid-ask quotes during period which sufficiently overlaps such an event is quite rare. For this reason, this study focuses on a specific/particular example for which I found reliable data. However, although I can provide reference to the source of this data (with thanks :)), I make no claim towards the accuracy of this data during the studied period.

- study type: Backtest of a strategy during a rare market event
- instrument: NVDA
- data source: option-chains: https://www.kaggle.com/datasets/kylegraupe/nvda-daily-option-chains-q1-2020-to-q4-2022
- data details: This dataset contain option-chains for NVDA, with bid-ask quotes during the period from 2 Jan 2020 to 29 December 2022. These quotes are for different expiry dates and strikes during this period.
- data frequency: The quotes are EOD (4pm Eastern) with daily frequency.
- analysis period
    - start-date: 2020-01-02
    - end-date: 2022-01-21
- other comments about the data & studied time-period: note that the options market for NVDA froze on 2021-07-21.

Goals & Objectives

The primary goal of this study is to define and backtest the strategy with the characteristics stated earlier. For achieving this, following sub-steps are needed.

- Data acquisition

- Define the strategy

- Pseudocode for implementing strategy

- Code the strategy

- Backtest the strategy during the period when the market freeze

- Present Results

References

- https://www.kaggle.com/datasets/kylegraupe/nvda-daily-option-chains-q1-2020-to-q4-2022